EXTROPY ENGINE · 6 SEPTEMBER 2026

Meter Math

How the meters couple. CT is community-web standing.
Reputation lives in CT. H starts at 0.5. EP dies in the sale.

Unsigned working spec for the tokenomics of the meters. Not bags. Nothing here cashes out.

House rule, first line: we do not play their markets. No offset registry. No transferable tonne. No Extropy credit listed next to Verra. If a room lists a spark as a commodity, they forked.

Signed Codex is still v2.1. Canonical mint in code is packages/xp-formula. If this page and the letter key disagree, the letter key(/key) wins on names. This page wins on how the meters couple.

Do not remap letters. Do not add a ninth instrument.

Words

- Tokenomics here means how XP, CT, H, L, EP, CAT, IT, leaks, gates, and CT-compatibility couple. Balancing acts.
- Token as a bag you transfer or cash out is a lose-condition.
- Spark is EP: born and burned in the sale.
- Community web W is the set of DFAOs that still read each other's base CT one-to-one.
- A chatbot that hears "tokenomics" and writes Uniswap is wrong.

What kind of object

Object	Kind	Liquid?	Who can change the rule
$R, F, \Delta S, w \cdot E, T_s$	Mint inputs	No	Planetary. Fork if you rewrite the XP mi
XP	Global standing meter	No. No cash-out.	Planetary. Earned on closed loops.
CT	Community standing meter on web W	No	DFAO posts task values and may wrap. Bas
ρ_W	Reputation density on web W	No	Enters CT only. Never the XP mint.
H	House slider	No. Vertex when it moves.	This till. Default 0.5. Auto recommended
κ	Compatibility of this door's CT with W	No	1 if base. (0,1) if a non-liquid wrap. 0
L	This-ticket standing	No	H, κ, CT_W, β.
EP	Till spark	Dies in the sale	EP = XP × L. Not payroll.
ρ	Fiat residual index	No	Door totals. Not a person. Not ρ_W.
CAT	Skill record	Unique	Off the XP mint.
IT	Governance meter	No	Off the XP mint.

0.99^n	XP leak	—	Planetary knob.
ℱ	Falsifiability	—	Lose-conditions. Not the letter F.

Cash still rings the rest of a ticket. Cash cannot mint XP. Cash-wrap of CT drops κ to 0.

1. The mint (signed)

Gates before the product is allowed to land:

- ΔS > 0 after the mapper and the leakage check.
- Both edges agreed.
- Rights / consent fail ⇒ ineligible. Not "big ΔS minus an ethics haircut."
- Unsettled vertices do not move IT.

XP is global. Every compatible room can see that a loop survived. Changing the XP mint is planetary. A DFAO that wants different local payouts uses CT, not a second XP.

Terms

R ∈ 0.1, 10 — rarity of the *action class*, not the person.

F — Frequency of Decay on that class. Not .

ΔS — bits-equivalent proxy. Not XP. Not SI social heat.

w · E — eight-domain weights dotted with this loop's effort.

T_s — slam window.

$T_s = \exp(-\lambda \min(\Delta t, \Delta t_{cap}))$

Δt → 0 ⇒ log = 0 ⇒ XP = 0. Three clocks: T_s this loop, F the class, 0.99^n standing after settle. IT leak is a fourth clock, on voice.

2. Mapper (constitution of ΔS)

domain-native signal M_v $\Delta S_{claimed}$ with uncertainty U on the vertex

$\Delta S = \Delta S_{claimed} - \Delta S_{leakage} - \Delta S_{displacement} - \Delta S_{unaccounted\ harm}$

Unknown does not default to 0. SignalFlow + the edge model + the PSLL propose. The other edge can refuse. Cross-reference on the DAG against similar timed events is how the proxy tightens. That is estimation. It is not a worshipped constant.

3. XP after it lands

No spend. Late mint / late burn are citation edges. Cash-out of XP is a lose-condition.

You can earn XP and CT on the same job. You can earn only one. See §4b.

4. CT — community standing on web W

CT is not “this one register.”
CT is standing in the community web of DFAOs that still share base CT.

Nodes: people.
Rooms: DFAOs.
Rooms attach to people and to each other.
CT answers: how useful was this node to *that web*, not only to one till.

Reputation belongs in CT. It was never supposed to leave. It does not belong in R of the XP mint.

4a. Working form

For web W, window voted by that web:

$U_W \cdot \rho_W \cdot C_W \cdot P_W \cdot (1 - F_W^{local})$
, 0, 1)

Symbol	Job
U_W	Usefulness minted from posted tasks this
ρ_W	Reputation density on W. Confirmed histo
C_W	Coupling across the web: closes / reliab
P_W	Predictability for the web. Random boom/
F_W^{local}	Farm penalty if the same class is being

A DFAO posts a task with a posted CT weight. SignalFlow proposes the close. Both edges agree. That close adds to U_W. Cross-ref similar DAG events so one shop cannot invent a private physics for “mow lawn = 900.”

CT is not purchased with XP. Dollars spent are not CT.

4b. Same job, two meters

- XP only — loop survived on the global mint. This web did not post it as a community task.
- CT only — this web posted it and paid CT. Mint gates for XP did not fire (or nobody filed ΔS).
- Both — posted task *and* a verified ΔS loop.
- Neither — cash only.

4c. Base, wrap, cash-wrap (compatibility κ)

Mode	What the DFAO did	κ	Still on XP network?
Base CT	Non-liquid community standing. Shared gr	1	Yes
Wrap	Local wrapper around CT (extra knobs), ex	$\in (0,1)$ as the web votes	Yes
Cash-wrap	Any exchange of CT for money, listed poi	0	XP may still exist. CT is no longer comp

A city-scale mesh that stays on base CT can treat standing as one-to-one across those rooms. Fork too far and it is not 1:1. Wire it to cash and it is not CT in this charter.

Axioms a wrap must keep to even *ask* for $\kappa > 0$:

- no liquidity
- no transfer-as-bag
- no cash-out
- reputation may stay in CT
- reputation stays out of the XP mint

4d. What the store does not own

The store does not own CT. The web does.
The store owns H.

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5. H — house slider

H ∈ 0, 1. Public. Moving it writes a vertex on this house.

Default 0.5. Sit in the middle so SignalFlow can auto-place H for this ticket (or this distributor-tier till) from applicable door totals and load. Auto is recommended. The house can always drag.

- H = 0 — overlay off. Allowed. Also a vertex. Parking it because you are "broke this month" still writes on *your* ledger. Hard times are real. The graph still records that you parked.
- H = 1 — maximum willingness to let compatible CT count against the published ticket cap.
- 0.5 — default. Fair-start. Auto has room to move.

Abuse (nuke H to starve a neighbor, juice H to puff your people, false-fraud shutdown) is a vertex on you. If you were wrong, that hits your own DAG, your own ρ as a door, your own next-fractal reading. Incentive not to be a dick is not a sermon. It is your books.

Distributor / B2B tills are still tills. Same H. Same spark. Same vertex.

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6. L — this ticket

$L = \text{clip}(H \cdot \kappa \cdot CT\_W \cdot \beta, 0, 1)$

β ∈ 0, 1 is domain bands shown to this door as ZKPs. No bands asked ⇒ β = 1.

Read it in English:
global standing is XP.
community standing is CT_W.
this house's willingness is H.
whether this house still speaks base CT is κ.
L is those things on this ticket.

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7. EP — till spark

Born and burned in the same sale. Not a wage. Not a currency. Not how staff or distributors get paid. Cash (or the local tender) still clears the invoice. The spark is an overlay. Early mesh: pennies. Dense compatible web: more of the ticket. Remainder rings in cash.

7b. Fiat residual (mesh index, not a person)

Door totals. Optional ZKP on the count. Not ρ_W. Different letter job.

7c. Carbon as a door, not a credit

Same mint. Same no-bag. Spark dies. We do not play their markets.

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8. CAT and IT (off the mint)

CAT — (DID, lane, level, issuer). Not a pile.

IT — voice.

Unsettled XP does not drive IT.

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9. Lose-conditions ()

- Cash-out of XP.
- Cash-wrap of CT (κ forced to 0; calling it “still base CT” is a lie).
- Silent rewrite of a neighbor’s DAG.
- Mapper that silently mutates history.
- Reputation smuggled into R of the XP mint.
- Rights / consent as a haircut instead of a gate.
- Mesh-wide forehead average of a person.
- Playing their carbon market.

A DFAO can fork CT. It does not get to keep $\kappa = 1$ if it wired money. It does not get to call a cash-wrap Codex v2.1.

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10. What this is not

Not speculative tokenomics.
Not six coins.
Not “CT is one store’s loyalty punch card.”

Not SI social entropy.

It is global standing, community standing on a compatible web, a house slider that starts at half, and one spark that dies when the ticket dies.

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  XP_mint = R × F × Δ S × (w · E) × log(1/T_s) XP(n) = XP_settled · 0.99ⁿ CT_W = clip( EP =
XP × L ρ = Σ_EP^$Σ_$ + Σ_EP^$ IT(m) = IT_idle-start · (0.95)ᵐ
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